Name: Deepali Paul

Project : Business Case: Netflix - Data Exploration and Visualisation

```
import numpy as np
import pandas as pd
import seaborn as sns
import matplotlib.pyplot as plt
```

```
!gdown 1XbDCjFt_LTTQGC6Mb9mCoVgpbR0VEMYx
```

```
Downloading...
From: https://drive.google.com/uc?id=1XbDCjFt_LTTQGC6Mb9mCoVgpbR0VEMYx
To: /content/netflix.csv
100% 3.40M/3.40M [00:00<00:00, 28.2MB/s]
```

```
df = pd.read_csv('netflix.csv')
```

```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 8807 entries, 0 to 8806
Data columns (total 12 columns):
 #   Column        Non-Null Count  Dtype 
---  ------        --------------  ----- 
 0   show_id       8807 non-null   object
 1   type          8807 non-null   object
 2   title         8807 non-null   object
 3   director      6173 non-null   object
 4   cast          7982 non-null   object
 5   country       7976 non-null   object
 6   date_added    8797 non-null   object
 7   release_year  8807 non-null   int64 
 8   rating        8803 non-null   object
 9   duration      8804 non-null   object
 10  listed_in     8807 non-null   object
 11  description   8807 non-null   object
dtypes: int64(1), object(11)
memory usage: 825.8+ KB
```

```
df.shape
```

```
(8807, 12)
```

--- ***The dataset[df] has 8807 rows and 12 columns***

```
df['date_added']= pd.to_datetime(df['date_added'], format='mixed')
df
```

	show_id	type	title	director	cast	country	date_added	release_year	rating	duration	listed_in	description
0	s1	Movie	Dick Johnson Is Dead	Kirsten Johnson	NaN	United States	2021-09-25	2020	PG-13	90 min	Documentaries	As her father nears the end of his life, filmm...
1	s2	TV Show	Blood & Water	NaN	Ama Qamata, Khosi Ngema, Gail Mabalane, Thaban...	South Africa	2021-09-24	2021	TV-MA	2 Seasons	International TV Shows, TV Dramas, TV Mysteries	After crossing paths at a party, a Cape Town t...
2	s3	TV Show	Ganglands	Julien Leclercq	Sami Bouajila, Tracy Gotoas, Samuel Jouy, Nabi...	NaN	2021-09-24	2021	TV-MA	1 Season	Crime TV Shows, International TV Shows, TV Act...	To protect his family from a powerful drug lor...
3	s4	TV Show	Jailbirds New Orleans	NaN	NaN	NaN	2021-09-24	2021	TV-MA	1 Season	Docuseries, Reality TV	Feuds, flirtations and toilet talk go down amo...
4	s5	TV Show	Kota Factory	NaN	Mayur More, Jitendra Kumar, Ranjan Raj, Alam K...	India	2021-09-24	2021	TV-MA	2 Seasons	International TV Shows, Romantic TV Shows, TV ...	In a city of coaching centers known to train I...
...	...	...	...	...	...	...	...	...	...	...	...	...
8802	s8803	Movie	Zodiac	David Fincher	Mark Ruffalo, Jake Gyllenhaal, Robert Downey J...	United States	2019-11-20	2007	R	158 min	Cult Movies, Dramas, Thrillers	A political cartoonist, a crime reporter and a...
8803	s8804	TV Show	Zombie Dumb	NaN	NaN	NaN	2019-07-01	2018	TV-Y7	2 Seasons	Kids' TV, Korean TV Shows, TV Comedies	While living alone in a spooky town, a young g...
8804	s8805	Movie	Zombieland	Ruben Fleischer	Jesse Eisenberg, Woody Harrelson,	United States	2019-11-01	2009	R	88 min	Comedies, Horror Movies	Looking to survive in a world taken

Next steps: Generate code with df | View recommended plots | New interactive sheet

Double-click (or enter) to edit

```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 8807 entries, 0 to 8806
Data columns (total 12 columns):
 #   Column        Non-Null Count  Dtype
---  ------        --------------  -----
 0   show_id       8807 non-null   object
 1   type          8807 non-null   object
 2   title         8807 non-null   object
 3   director      6173 non-null   object
 4   cast          7982 non-null   object
 5   country       7976 non-null   object
 6   date_added    8797 non-null   datetime64[ns]
 7   release_year  8807 non-null   int64
 8   rating        8803 non-null   object
 9   duration      8804 non-null   object
 10  listed_in     8807 non-null   object
 11  description   8807 non-null   object
dtypes: datetime64[ns](1), int64(1), object(10)
memory usage: 825.8+ KB
```

```
#adding week to the table

df['week']= df['date_added'].dt.isocalendar().week
df
```

	show_id	type	title	director	cast	country	date_added	release_year	rating	duration	listed_in	description	we
0	s1	Movie	Dick Johnson Is Dead	Kirsten Johnson	NaN	United States	2021-09-25	2020	PG-13	90 min	Documentaries	As her father nears the end of his life, filmm...	3
1	s2	TV Show	Blood & Water	NaN	Ama Qamata, Khosi Ngema, Gail Mabalane, Thaban...	South Africa	2021-09-24	2021	TV-MA	2 Seasons	International TV Shows, TV Dramas, TV Mysteries	After crossing paths at a party, a Cape Town t...	3
2	s3	TV Show	Ganglands	Julien Leclercq	Sami Bouajila, Tracy Gotoas, Samuel Jouy, Nabi...	NaN	2021-09-24	2021	TV-MA	1 Season	Crime TV Shows, International TV Shows, TV Act...	To protect his family from a powerful drug lor...	3
3	s4	TV Show	Jailbirds New Orleans	NaN	NaN	NaN	2021-09-24	2021	TV-MA	1 Season	Docuseries, Reality TV	Feuds, flirtations and toilet talk go down amo...	3
4	s5	TV Show	Kota Factory	NaN	Mayur More, Jitendra Kumar, Ranjan Raj, Alam K...	India	2021-09-24	2021	TV-MA	2 Seasons	International TV Shows, Romantic TV Shows, TV ...	In a city of coaching centers known to train I...	3
...	...	...	...	...	...	...	...	...	...	...	...	...	
8802	s8803	Movie	Zodiac	David Fincher	Mark Ruffalo, Jake Gyllenhaal, Robert Downey J...	United States	2019-11-20	2007	R	158 min	Cult Movies, Dramas, Thrillers	A political cartoonist, a crime reporter and a...	4
8803	s8804	TV Show	Zombie Dumb	NaN	NaN	NaN	2019-07-01	2018	TV-Y7	2 Seasons	Kids' TV, Korean TV Shows, TV Comedies	While living alone in a spooky town, a young g...	2
8804	s8805	Movie	Zombieland	Ruben Fleischer	Jesse Eisenberg, Woody Harrelson, Emma Stone, ...	United States	2019-11-01	2009	R	88 min	Comedies, Horror Movies	Looking to survive in a world taken over by zo...	4
8805	s8806	Movie	Zoom	Peter Hewitt	Tim Allen, Courteney Cox, Chevy Chase, Kate Ma...	United States	2020-01-11	2006	PG	88 min	Children & Family Movies, Comedies	Dragged from civilian life, a former superhero...	

Next steps: Generate code with df | View recommended plots | New interactive sheet

```
#add year number
df['year']= df['date_added'].apply(lambda x: x.year)

df
```

	show_id	type	title	director	cast	country	date_added	release_year	rating	duration	listed_in	description	we
0	s1	Movie	Dick Johnson Is Dead	Kirsten Johnson	NaN	United States	2021-09-25	2020	PG-13	90 min	Documentaries	As her father nears the end of his life, filmm...	
1	s2	TV Show	Blood & Water	NaN	Ama Qamata, Khosi Ngema, Gail Mabalane, Thaban...	South Africa	2021-09-24	2021	TV-MA	2 Seasons	International TV Shows, TV Dramas, TV Mysteries	After crossing paths at a party, a Cape Town t...	
2	s3	TV Show	Ganglands	Julien Leclercq	Sami Bouajila, Tracy Gotoas, Samuel Jouy, Nabi...	NaN	2021-09-24	2021	TV-MA	1 Season	Crime TV Shows, International TV Shows, TV Act...	To protect his family from a powerful drug lor...	
3	s4	TV Show	Jailbirds New Orleans	NaN	NaN	NaN	2021-09-24	2021	TV-MA	1 Season	Docuseries, Reality TV	Feuds, flirtations and toilet talk go down amo...	
4	s5	TV Show	Kota Factory	NaN	Mayur More, Jitendra Kumar, Ranjan Raj, Alam K...	India	2021-09-24	2021	TV-MA	2 Seasons	International TV Shows, Romantic TV Shows, TV ...	In a city of coaching centers known to train I...	
...	...	...	...	...	...	...	...	...	...	...	...	...	
8802	s8803	Movie	Zodiac	David Fincher	Mark Ruffalo, Jake Gyllenhaal, Robert Downey J...	United States	2019-11-20	2007	R	158 min	Cult Movies, Dramas, Thrillers	A political cartoonist, a crime reporter and a...	
8803	s8804	TV Show	Zombie Dumb	NaN	NaN	NaN	2019-07-01	2018	TV-Y7	2 Seasons	Kids' TV, Korean TV Shows, TV Comedies	While living alone in a spooky town, a young g...	
8804	s8805	Movie	Zombieland	Ruben Fleischer	Jesse Eisenberg, Woody Harrelson, Emma Stone, ...	United States	2019-11-01	2009	R	88 min	Comedies, Horror Movies	Looking to survive in a world taken over by zo...	
8805	s8806	Movie	Zoom	Peter Hewitt	Tim Allen, Courteney Cox, Chevy Chase, Kate Ma...	United States	2020-01-11	2006	PG	88 min	Children & Family Movies, Comedies	Dragged from civilian life, a former superhero...	
8806	s8807	Movie	Zubaan	Mozez Singh	Vicky Kaushal, Sarah-Jane Dias, Raaghav Chanan...	India	2019-03-02	2015	TV-14	111 min	Dramas, International Movies, Music & Musicals	A scrappy but poor boy worms his way into a ty...	

8807 rows × 14 columns

Next steps: Generate code with df | View recommended plots | New interactive sheet

```
df['year'] = df['year'].fillna(value = 0)
df
```

	show_id	type	title	director	cast	country	date_added	release_year	rating	duration	listed_in	description	we
0	s1	Movie	Dick Johnson Is Dead	Kirsten Johnson	NaN	United States	2021-09-25	2020	PG-13	90 min	Documentaries	As her father nears the end of his life, filmm...	
1	s2	TV Show	Blood & Water	NaN	Ama Qamata, Khosi Ngema, Gail Mabalane, Thaban...	South Africa	2021-09-24	2021	TV-MA	2 Seasons	International TV Shows, TV Dramas, TV Mysteries	After crossing paths at a party, a Cape Town t...	
2	s3	TV Show	Ganglands	Julien Leclercq	Sami Bouajila, Tracy Gotoas, Samuel Jouy, Nabi...	NaN	2021-09-24	2021	TV-MA	1 Season	Crime TV Shows, International TV Shows, TV Act...	To protect his family from a powerful drug lor...	
3	s4	TV Show	Jailbirds New Orleans	NaN	NaN	NaN	2021-09-24	2021	TV-MA	1 Season	Docuseries, Reality TV	Feuds, flirtations and toilet talk go down amo...	
4	s5	TV Show	Kota Factory	NaN	Mayur More, Jitendra Kumar, Ranjan Raj, Alam K...	India	2021-09-24	2021	TV-MA	2 Seasons	International TV Shows, Romantic TV Shows, TV ...	In a city of coaching centers known to train I...	
...	...	...	...	...	...	...	...	...	...	...	...	...	
8802	s8803	Movie	Zodiac	David Fincher	Mark Ruffalo, Jake Gyllenhaal, Robert Downey J...	United States	2019-11-20	2007	R	158 min	Cult Movies, Dramas, Thrillers	A political cartoonist, a crime reporter and a...	
8803	s8804	TV Show	Zombie Dumb	NaN	NaN	NaN	2019-07-01	2018	TV-Y7	2 Seasons	Kids' TV, Korean TV Shows, TV Comedies	While living alone in a spooky town, a young g...	
8804	s8805	Movie	Zombieland	Ruben Fleischer	Jesse Eisenberg, Woody Harrelson, Emma Stone, ...	United States	2019-11-01	2009	R	88 min	Comedies, Horror Movies	Looking to survive in a world taken over by zo...	
8805	s8806	Movie	Zoom	Peter Hewitt	Tim Allen, Courteney Cox, Chevy Chase, Kate Ma...	United States	2020-01-11	2006	PG	88 min	Children & Family Movies, Comedies	Dragged from civilian life, a former superhero...	
8806	s8807	Movie	Zubaan	Mozez Singh	Vicky Kaushal, Sarah-Jane Dias, Raaghav Chanan...	India	2019-03-02	2015	TV-14	111 min	Dramas, International Movies, Music & Musicals	A scrappy but poor boy worms his way into a ty...	

8807 rows × 14 columns

Next steps: Generate code with df | View recommended plots | New interactive sheet

```
df['year'] = df['year'].astype(int)
df
```

	show_id	type	title	director	cast	country	date_added	release_year	rating	duration	listed_in	description	we
0	s1	Movie	Dick Johnson Is Dead	Kirsten Johnson	NaN	United States	2021-09-25	2020	PG-13	90 min	Documentaries	As her father nears the end of his life, filmm...	
1	s2	TV Show	Blood & Water	NaN	Ama Qamata, Khosi Ngema, Gail Mabalane, Thaban...	South Africa	2021-09-24	2021	TV-MA	2 Seasons	International TV Shows, TV Dramas, TV Mysteries	After crossing paths at a party, a Cape Town t...	
2	s3	TV Show	Ganglands	Julien Leclercq	Sami Bouajila, Tracy Gotoas, Samuel Jouy, Nabi...	NaN	2021-09-24	2021	TV-MA	1 Season	Crime TV Shows, International TV Shows, TV Act...	To protect his family from a powerful drug lor...	
3	s4	TV Show	Jailbirds New Orleans	NaN	NaN	NaN	2021-09-24	2021	TV-MA	1 Season	Docuseries, Reality TV	Feuds, flirtations and toilet talk go down amo...	
4	s5	TV Show	Kota Factory	NaN	Mayur More, Jitendra Kumar, Ranjan Raj, Alam K...	India	2021-09-24	2021	TV-MA	2 Seasons	International TV Shows, Romantic TV Shows, TV ...	In a city of coaching centers known to train I...	
...	...	...	...	...	...	...	...	...	...	...	...	...	
8802	s8803	Movie	Zodiac	David Fincher	Mark Ruffalo, Jake Gyllenhaal, Robert Downey J...	United States	2019-11-20	2007	R	158 min	Cult Movies, Dramas, Thrillers	A political cartoonist, a crime reporter and a...	
8803	s8804	TV Show	Zombie Dumb	NaN	NaN	NaN	2019-07-01	2018	TV-Y7	2 Seasons	Kids' TV, Korean TV Shows, TV Comedies	While living alone in a spooky town, a young g...	
8804	s8805	Movie	Zombieland	Ruben Fleischer	Jesse Eisenberg, Woody Harrelson, Emma Stone, ...	United States	2019-11-01	2009	R	88 min	Comedies, Horror Movies	Looking to survive in a world taken over by zo...	
8805	s8806	Movie	Zoom	Peter Hewitt	Tim Allen, Courteney Cox, Chevy Chase, Kate Ma...	United States	2020-01-11	2006	PG	88 min	Children & Family Movies, Comedies	Dragged from civilian life, a former superhero...	
8806	s8807	Movie	Zubaan	Mozez Singh	Vicky Kaushal, Sarah-Jane Dias, Raaghav Chanan...	India	2019-03-02	2015	TV-14	111 min	Dramas, International Movies, Music & Musicals	A scrappy but poor boy worms his way into a ty...	

8807 rows × 14 columns

Next steps: Generate code with df | View recommended plots | New interactive sheet

```
#add month number
df['month']= df['date_added'].dt.month_name()
df
```

	show_id	type	title	director	cast	country	date_added	release_year	rating	duration	listed_in	description	we
0	s1	Movie	Dick Johnson Is Dead	Kirsten Johnson	NaN	United States	2021-09-25	2020	PG-13	90 min	Documentaries	As her father nears the end of his life, filmm...	
1	s2	TV Show	Blood & Water	NaN	Ama Qamata, Khosi Ngema, Gail Mabalane, Thaban...	South Africa	2021-09-24	2021	TV-MA	2 Seasons	International TV Shows, TV Dramas, TV Mysteries	After crossing paths at a party, a Cape Town t...	
2	s3	TV Show	Ganglands	Julien Leclercq	Sami Bouajila, Tracy Gotoas, Samuel Jouy, Nabi...	NaN	2021-09-24	2021	TV-MA	1 Season	Crime TV Shows, International TV Shows, TV Act...	To protect his family from a powerful drug lor...	
3	s4	TV Show	Jailbirds New Orleans	NaN	NaN	NaN	2021-09-24	2021	TV-MA	1 Season	Docuseries, Reality TV	Feuds, flirtations and toilet talk go down amo...	
4	s5	TV Show	Kota Factory	NaN	Mayur More, Jitendra Kumar, Ranjan Raj, Alam K...	India	2021-09-24	2021	TV-MA	2 Seasons	International TV Shows, Romantic TV Shows, TV ...	In a city of coaching centers known to train I...	
...	...	...	...	...	...	...	...	...	...	...	...	...	
8802	s8803	Movie	Zodiac	David Fincher	Mark Ruffalo, Jake Gyllenhaal, Robert Downey J...	United States	2019-11-20	2007	R	158 min	Cult Movies, Dramas, Thrillers	A political cartoonist, a crime reporter and a...	
8803	s8804	TV Show	Zombie Dumb	NaN	NaN	NaN	2019-07-01	2018	TV-Y7	2 Seasons	Kids' TV, Korean TV Shows, TV Comedies	While living alone in a spooky town, a young g...	
8804	s8805	Movie	Zombieland	Ruben Fleischer	Jesse Eisenberg, Woody Harrelson, Emma Stone, ...	United States	2019-11-01	2009	R	88 min	Comedies, Horror Movies	Looking to survive in a world taken over by zo...	
8805	s8806	Movie	Zoom	Peter Hewitt	Tim Allen, Courteney Cox, Chevy Chase, Kate Ma...	United States	2020-01-11	2006	PG	88 min	Children & Family Movies, Comedies	Dragged from civilian life, a former superhero...	
8806	s8807	Movie	Zubaan	Mozez Singh	Vicky Kaushal, Sarah-Jane Dias, Raaghav Chanan...	India	2019-03-02	2015	TV-14	111 min	Dramas, International Movies, Music & Musicals	A scrappy but poor boy worms his way into a ty...	

8807 rows × 15 columns

Next steps: Generate code with df | View recommended plots | New interactive sheet

```
#Handling null values
value = {'director':'unknown director', 'cast':'unknown cast', 'country':'unknown country', 'date_added':'unknown date_added', 'rating':'unkn
df = df.fillna(value = value)
df
```

	show_id	type	title	director	cast	country	date_added	release_year	rating	duration	listed_in	description	we
0	s1	Movie	Dick Johnson Is Dead	Kirsten Johnson	unknown cast	United States	2021-09-25 00:00:00	2020	PG-13	90 min	Documentaries	As her father nears the end of his life, filmm...	
1	s2	TV Show	Blood & Water	unknown director	Ama Qamata, Khosi Ngema, Gail Mabalane, Thaban...	South Africa	2021-09-24 00:00:00	2021	TV-MA	2 Seasons	International TV Shows, TV Dramas, TV Mysteries	After crossing paths at a party, a Cape Town t...	
2	s3	TV Show	Ganglands	Julien Leclercq	Sami Bouajila, Tracy Gotoas, Samuel Jouy, Nabi...	unknown country	2021-09-24 00:00:00	2021	TV-MA	1 Season	Crime TV Shows, International TV Shows, TV Act...	To protect his family from a powerful drug lor...	
3	s4	TV Show	Jailbirds New Orleans	unknown director	unknown cast	unknown country	2021-09-24 00:00:00	2021	TV-MA	1 Season	Docuseries, Reality TV	Feuds, flirtations and toilet talk go down amo...	
4	s5	TV Show	Kota Factory	unknown director	Mayur More, Jitendra Kumar, Ranjan Raj, Alam K...	India	2021-09-24 00:00:00	2021	TV-MA	2 Seasons	International TV Shows, Romantic TV Shows, TV ...	In a city of coaching centers known to train I...	
...	...	...	...	...	...	...	...	...	...	...	...	...	
8802	s8803	Movie	Zodiac	David Fincher	Mark Ruffalo, Jake Gyllenhaal, Robert Downey J...	United States	2019-11-20 00:00:00	2007	R	158 min	Cult Movies, Dramas, Thrillers	A political cartoonist, a crime reporter and a...	
8803	s8804	TV Show	Zombie Dumb	unknown director	unknown cast	unknown country	2019-07-01 00:00:00	2018	TV-Y7	2 Seasons	Kids' TV, Korean TV Shows, TV Comedies	While living alone in a spooky town, a young g...	
8804	s8805	Movie	Zombieland	Ruben Fleischer	Jesse Eisenberg, Woody Harrelson, Emma Stone, ...	United States	2019-11-01 00:00:00	2009	R	88 min	Comedies, Horror Movies	Looking to survive in a world taken over by zo...	
8805	s8806	Movie	Zoom	Peter Hewitt	Tim Allen, Courteney Cox, Chevy Chase, Kate Ma...	United States	2020-01-11 00:00:00	2006	PG	88 min	Children & Family Movies, Comedies	Dragged from civilian life, a former superhero...	
8806	s8807	Movie	Zubaan	Mozez Singh	Vicky Kaushal, Sarah-Jane Dias, Raaghav Chanan...	India	2019-03-02 00:00:00	2015	TV-14	111 min	Dramas, International Movies, Music & Musicals	A scrappy but poor boy worms his way into a ty...	

8807 rows × 15 columns

Next steps: Generate code with df | View recommended plots | New interactive sheet

```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 8807 entries, 0 to 8806
Data columns (total 15 columns):
 #   Column        Non-Null Count  Dtype
---  ------        --------------  -----
```

```
 0   show_id       8807 non-null   object
 1   type          8807 non-null   object
 2   title         8807 non-null   object
 3   director      8807 non-null   object
 4   cast          8807 non-null   object
 5   country       8807 non-null   object
 6   date_added    8807 non-null   object
 7   release_year  8807 non-null   int64
 8   rating        8807 non-null   object
 9   duration      8807 non-null   object
 10  listed_in     8807 non-null   object
 11  description   8807 non-null   object
 12  week          8797 non-null   UInt32
 13  year          8807 non-null   int64
 14  month         8797 non-null   object
dtypes: UInt32(1), int64(2), object(12)
memory usage: 1006.4+ KB
```

```
#movies table
df_movies = df[df['type'] == 'Movie']
df_movies
```

	show_id	type	title	director	cast	country	date_added	release_year	rating	duration	listed_in	description
0	s1	Movie	Dick Johnson Is Dead	Kirsten Johnson	unknown cast	United States	2021-09-25 00:00:00	2020	PG-13	90 min	Documentaries	As her father nears the end of his life, filmm...
6	s7	Movie	My Little Pony: A New Generation	Robert Cullen, José Luis Ucha	Vanessa Hudgens, Kimiko Glenn, James Marsden, ...	unknown country	2021-09-24 00:00:00	2021	PG	91 min	Children & Family Movies	Equestria's divided. But a bright-eyed hero be...
7	s8	Movie	Sankofa	Haile Gerima	Kofi Ghanaba, Oyafunmike Ogunlano, Alexandra D...	United States, Ghana, Burkina Faso, United Kin...	2021-09-24 00:00:00	1993	TV-MA	125 min	Dramas, Independent Movies, International Movies	On a photo shoot in Ghana, an American model s...
9	s10	Movie	The Starling	Theodore Melfi	Melissa McCarthy, Chris O'Dowd, Kevin Kline, T...	United States	2021-09-24 00:00:00	2021	PG-13	104 min	Comedies, Dramas	A woman adjusting to life after a loss contend...
12	s13	Movie	Je Suis Karl	Christian Schwochow	Luna Wedler, Jannis Niewöhner, Milan Peschel, ...	Germany, Czech Republic	2021-09-23 00:00:00	2021	TV-MA	127 min	Dramas, International Movies	After most of her family is murdered in a terr...
...	...	...	...	...	...	...	...	...	...	...	...	...
8801	s8802	Movie	Zinzana	Majid Al Ansari	Ali Suliman, Saleh Bakri, Yasa, Ali Al-Jabri, ...	United Arab Emirates, Jordan	2016-03-09 00:00:00	2015	TV-MA	96 min	Dramas, International Movies, Thrillers	Recovering alcoholic Talal wakes up inside a s...
8802	s8803	Movie	Zodiac	David Fincher	Mark Ruffalo, Jake Gyllenhaal, Robert Downey J...	United States	2019-11-20 00:00:00	2007	R	158 min	Cult Movies, Dramas, Thrillers	A political cartoonist, a crime reporter and a...
8804	s8805	Movie	Zombieland	Ruben Fleischer	Jesse Eisenberg, Woody Harrelson, Emma Stone, ...	United States	2019-11-01 00:00:00	2009	R	88 min	Comedies, Horror Movies	Looking to survive in a world taken over by zo...
8805	s8806	Movie	Zoom	Peter Hewitt	Tim Allen, Courteney Cox, Chevy Chase, Kate Ma...	United States	2020-01-11 00:00:00	2006	PG	88 min	Children & Family Movies, Comedies	Dragged from civilian life, a former superhero...
8806	s8807	Movie	Zubaan	Mozez Singh	Vicky Kaushal, Sarah-Jane Dias, Raaghav Chanan...	India	2019-03-02 00:00:00	2015	TV-14	111 min	Dramas, International Movies, Music & Musicals	A scrappy but poor boy worms his way into a ty...

6131 rows × 15 columns

Next steps: Generate code with df_movies | View recommended plots | New interactive sheet

```
#tv shows table
df_tvshows = df[df['type'] == 'TV Show']
df_tvshows
```

	show_id	type	title	director	cast	country	date_added	release_year	rating	duration	listed_in	description	week
1	s2	TV Show	Blood & Water	unknown director	Ama Qamata, Khosi Ngema, Gail Mabalane, Thaban...	South Africa	2021-09-24 00:00:00	2021	TV-MA	2 Seasons	International TV Shows, TV Dramas, TV Mysteries	After crossing paths at a party, a Cape Town t...	38
2	s3	TV Show	Ganglands	Julien Leclercq	Sami Bouajila, Tracy Gotoas, Samuel Jouy, Nabi...	unknown country	2021-09-24 00:00:00	2021	TV-MA	1 Season	Crime TV Shows, International TV Shows, TV Act...	To protect his family from a powerful drug lor...	38
3	s4	TV Show	Jailbirds New Orleans	unknown director	unknown cast	unknown country	2021-09-24 00:00:00	2021	TV-MA	1 Season	Docuseries, Reality TV	Feuds, flirtations and toilet talk go down amo...	38
4	s5	TV Show	Kota Factory	unknown director	Mayur More, Jitendra Kumar, Ranjan Raj, Alam K...	India	2021-09-24 00:00:00	2021	TV-MA	2 Seasons	International TV Shows, Romantic TV Shows, TV ...	In a city of coaching centers known to train I...	38
5	s6	TV Show	Midnight Mass	Mike Flanagan	Kate Siegel, Zach Gilford, Hamish Linklater, H...	unknown country	2021-09-24 00:00:00	2021	TV-MA	1 Season	TV Dramas, TV Horror, TV Mysteries	The arrival of a charismatic young priest brin...	38
...	...	...	...	...	...	...	...	...	...	...	...	...	...
8795	s8796	TV Show	Yu-Gi-Oh! Arc-V	unknown director	Mike Liscio, Emily Bauer, Billy Bob Thompson, ...	Japan, Canada	2018-05-01 00:00:00	2015	TV-Y7	2 Seasons	Anime Series, Kids' TV	Now that he's discovered the Pendulum Summonin...	18
8796	s8797	TV Show	Yunus Emre	unknown director	Gökhan Atalay, Payidar Tüfekçioglu, Baran Akbu...	Turkey	2017-01-17 00:00:00	2016	TV-PG	2 Seasons	International TV Shows, TV Dramas	During the Mongol invasions, Yunus Emre leaves...	3
8797	s8798	TV Show	Zak Storm	unknown director	Michael Johnston, Jessica Gee-George, Christin...	United States, France, South Korea, Indonesia	2018-09-13 00:00:00	2016	TV-Y7	3 Seasons	Kids' TV	Teen surfer Zak Storm is mysteriously transpor...	37
8800	s8801	TV Show	Zindagi Gulzar Hai	unknown director	Sanam Saeed, Fawad Khan, Ayesha Omer, Mehreen ...	Pakistan	2016-12-15 00:00:00	2012	TV-PG	1 Season	International TV Shows, Romantic TV Shows, TV ...	Strong-willed, middle-class Kashaf and carefre...	50
8803	s8804	TV Show	Zombie Dumb	unknown director	unknown cast	unknown country	2019-07-01 00:00:00	2018	TV-Y7	2 Seasons	Kids' TV, Korean TV Shows, TV Comedies	While living alone in a spooky town, a young g...	27

2676 rows × 15 columns

Next steps: Generate code with df_tvshows | View recommended plots | New interactive sheet

```
#Find the number of movies produced in each country and pick the top 10
df_movies.groupby('country')['title'].count().sort_values(ascending=False).head(10)
```

country	title
United States	2058
India	893
unknown country	440
United Kingdom	206
Canada	122
Spain	97
Egypt	92
Nigeria	86
Indonesia	77
Turkey	76

dtype: int64

--- **Top most movies have been produced by Unites states, followed by India and United Kingdom.**

```
#Find the number of Tv-Shows produced in each country and pick the top 10
df_tvshows.groupby('country')['title'].count().sort_values(ascending=False).head(10)
```

country	title
United States	760
unknown country	391
United Kingdom	213
Japan	169
South Korea	158
India	79
Taiwan	68
Canada	59
France	49
Spain	48

dtype: int64

--- **Top most Tv shows has been produced by United States, United Kingdon and Japan**

It means --- **US and UK has prduced most number of tv shows and movies. countries like India and Canada produce large number of movies but not TV shows - on the other hand Japan and South Korea has huge number of TV shows produced in Netflix.**

```
#The best week to release a movie.
df_movies.groupby('week')['title'].count().sort_values(ascending=False).head(1)
```

week	title
1	316

dtype: int64

```
#The best week to release a Tv-show
df_tvshows.groupby('week')['title'].count().sort_values(ascending=False).head(1)
```

week	title
27	86

dtype: int64

```
#The best month to release a movie
df_movies.groupby('month')['title'].count().sort_values(ascending=False).head(1)
```

month	title
July	565

dtype: int64

```
#The best month to release a Tv-show
df_tvshows.groupby('month')['title'].count().sort_values(ascending=False).head(1)
```

month	title
December	266

dtype: int64

```
#Identify the top 10 directors who have appeared in most TV shows.
director_movies = df_movies[['director']].apply(lambda x: x.str.split(',')).explode(['director']).reset_index(drop=True)
director_movies.value_counts().head(10)
```

director	count
unknown director	188
Rajiv Chilaka	22
Jan Suter	18
Raúl Campos	18
Suhas Kadav	16
Jay Karas	15
Marcus Raboy	15
Cathy Garcia-Molina	13
Martin Scorsese	12
Youssef Chahine	12

dtype: int64

--- **Even though most of the data is unknown, it is quite clear then Rajib Chilaka has the highest contribution as a director**

```
#Identify the top 10 directors who have appeared in most TV show
director_tvshows = df_tvshows[['director']].apply(lambda x: x.str.split(',')).explode(['director']).reset_index(drop=True)
director_tvshows.value_counts().head(10)
```

director	count
unknown director	2446
Ken Burns	3
Alastair Fothergill	3
Gautham Vasudev Menon	2
Iginio Straffi	2
Joe Berlinger	2
Jung-ah Im	2
Rob Seidenglanz	2
Shin Won-ho	2
Stan Lathan	2

dtype: int64

--- **Similarly in TV shows Ken Burns and Alastair Fothergill has directed hight amount - however, true data can not be found since most of the data is unknown**

```
#Identify the top 10 actor who have appeared in most movies.
cast_movies = df_movies[['cast']].apply(lambda x: x.str.split(',')).explode(['cast']).reset_index(drop=True)
cast_movies.value_counts().head(10)
```

cast	count
unknown cast	475
Anupam Kher	38
Rupa Bhimani	27
Om Puri	27
Shah Rukh Khan	26
Paresh Rawal	25
Boman Irani	25
Julie Tejwani	24
Akshay Kumar	23
Rajesh Kava	21

dtype: int64

----- ***Anupam Kher, Rupa Bhimani, Om Puri, Shah Rukh Khan are the are the actors who has been filmed in most number of movies***

```
#Identify the top 10 actor who have appeared in most TV shows.
cast_tvshows = df_tvshows[['cast']].apply(lambda x: x.str.split(',')).explode(['cast']).reset_index(drop=True)
cast_tvshows.value_counts().head(10)
```

cast	count
unknown cast	350
Takahiro Sakurai	24
Yuki Kaji	17
Junichi Suwabe	17
Ai Kayano	17
Daisuke Ono	14
David Attenborough	14
Takehito Koyasu	13
Yoshimasa Hosoya	13
Yuichi Nakamura	13

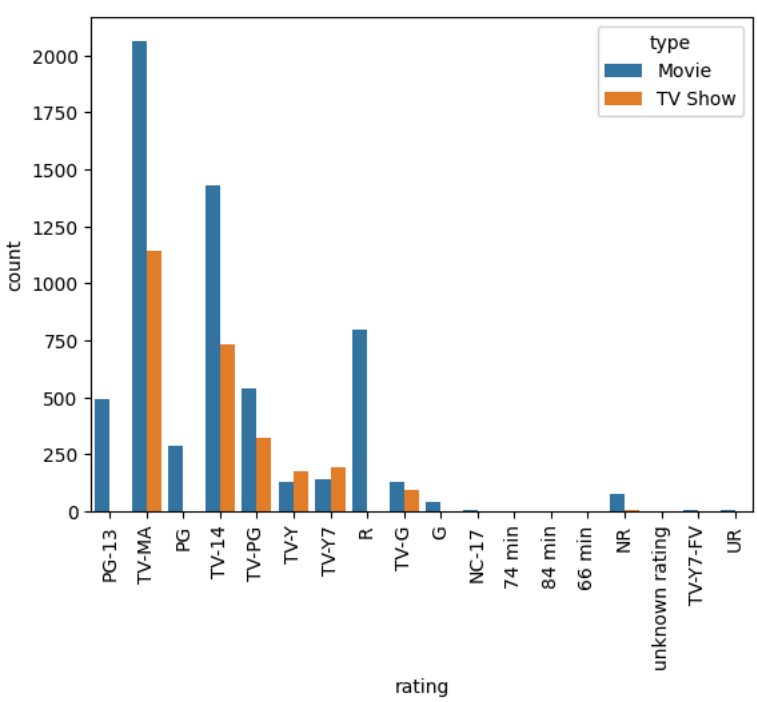
dtype: int64

--- ***Takahiro Sakurai has been filmed in 24 movies***

```
#Which genre movies are more popular or produced more
# - > we do not have genre column hence using rating column

popular_genre = df_movies['listed_in'].value_counts().head(10)
```

```
sns.countplot(data = df , x = 'rating', hue = 'type')
plt.xticks(rotation=90)
plt.show()
```



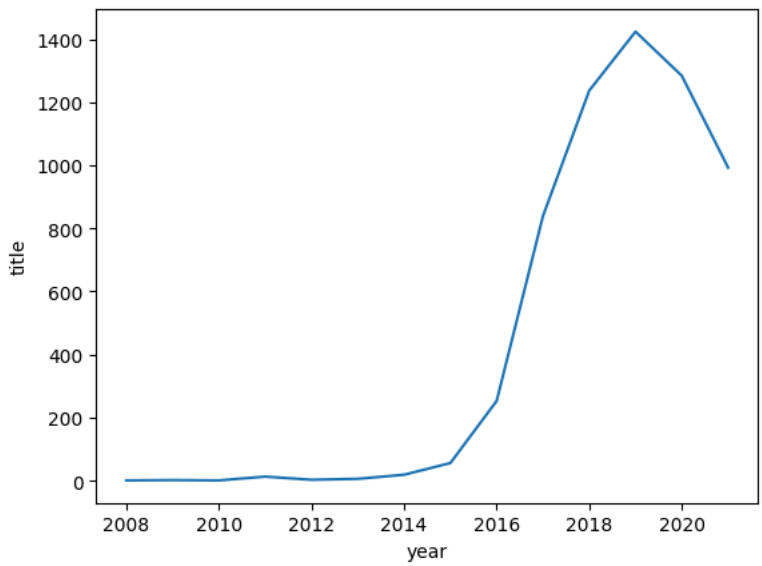
--- ***TV-MA and TV-14 rated TV shows or movies have been produced then most***

```
movie_per_year = df_movies.groupby('year')['title'].count().reset_index()
movie_per_year.sort_values('year', ascending=False)
```

	year	title
13	2021	993
12	2020	1284
11	2019	1424
10	2018	1237
9	2017	839
8	2016	253
7	2015	56
6	2014	19
5	2013	6
4	2012	3
3	2011	13
2	2010	1
1	2009	2
0	2008	1

```
sns.lineplot(data = movie_per_year, x= 'year', y='title')
```

```
<Axes: xlabel='year', ylabel='title'>
```



--- Movie produced has a growing trend over year and it has touched peak in 2019 with 1424 movies. However, it has drop a lot in the next 2 years

```
tvshows_per_year = df_tvshows.groupby('year')['title'].count().reset_index()
tvshows_per_year.sort_values('year', ascending=False)
```

	year	title
10	2021	505
9	2020	595
8	2019	592
7	2018	412
6	2017	349
5	2016	176
4	2015	26
3	2014	5
2	2013	5
1	2008	1
0	0	10

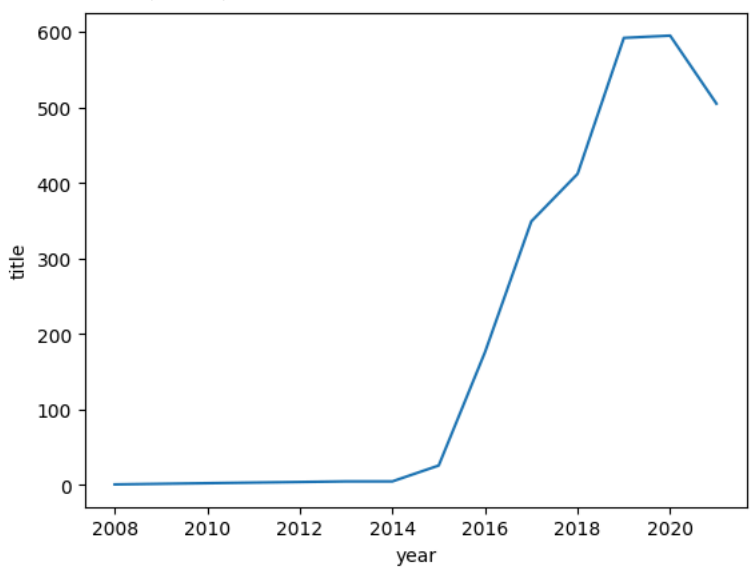
```
tvshows_per_year.drop(0,axis=0, inplace= True)
tvshows_per_year
```

	year	title
1	2008	1
2	2013	5
3	2014	5
4	2015	26
5	2016	176
6	2017	349
7	2018	412
8	2019	592
9	2020	595
10	2021	505

Next steps: Generate code with tvshows_per_year | View recommended plots | New interactive sheet

```
sns.lineplot(data = tvshows_per_year, x= 'year', y='title')
```

```
<Axes: xlabel='year', ylabel='title'>
```



--- TV shows have been in a growing trend. in the year 2019-2020 netflix has produced hight amount of shows. in the year 2021 the number has fallen down

```
duration = df.groupby(['type','duration'])['title'].count().reset_index()
duration.sort_values('title', ascending=False).head(20)
```

	type	duration	title
206	TV Show	1 Season	1793
213	TV Show	2 Seasons	425
214	TV Show	3 Seasons	199
195	Movie	90 min	152
202	Movie	97 min	146
198	Movie	93 min	146
199	Movie	94 min	146
196	Movie	91 min	144
200	Movie	95 min	137
201	Movie	96 min	130
197	Movie	92 min	129
3	Movie	102 min	122
203	Movie	98 min	120
204	Movie	99 min	118
2	Movie	101 min	116
192	Movie	88 min	116
4	Movie	103 min	114
7	Movie	106 min	111
1	Movie	100 min	108
193	Movie	89 min	106

--- We can conclude that shorter-duration content has been in demand over the years. movies with 1 season has been added in a huge amount tv-shows within [90 mins to 105 mins] have beem largely produced

Start coding or generate with AI.

Start coding or generate with AI.

Start coding or generate with AI.

Start coding or generate with AI.

Start coding or generate with AI.

Start coding or generate with AI.

Start coding or generate with AI.

Start coding or generate with AI.

Start coding or generate with AI.

Start coding or generate with AI.

Start coding or generate with AI.

Start coding or generate with AI.

Start coding or generate with AI.